

Auteur: Siebe Mees
 Studierichting: Informatica
 Oefeningenreeks 5F nr. 1.4

$$r = 9 \sin 2\theta$$

$$r' = 18 \cos 2\theta$$

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
r	0	9	0	-9	0
r'	+	0	-	0	+
r	$\nearrow$		$\searrow$		$\nearrow$

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$$\begin{aligned}
 S_1 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (9 \sin 2\theta)^2 d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} 81 \sin^2 2\theta d\theta \\
 &= \frac{81}{2} \int_0^{\frac{\pi}{2}} \sin^2 2\theta d\theta \\
 &= \frac{81}{2} \int_0^{\frac{\pi}{2}} \frac{1 - \cos 4\theta}{2} d\theta \\
 &= \frac{81}{4} \int_0^{\frac{\pi}{2}} (1 - \cos 4\theta) d\theta \\
 &= \frac{81}{4} \left[\theta - \frac{1}{4} \sin 4\theta \right]_0^{\frac{\pi}{2}} \\
 &= \frac{81}{4} \left(\frac{\pi}{2} - 0 \right) \\
 &= \frac{81\pi}{8} \\
 S &= 4 \cdot S_1
 \end{aligned}$$

$$\begin{aligned}
&= 4 \cdot \frac{81\pi}{8} \\
&= \frac{81\pi}{2}
\end{aligned}$$

References